Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

FOOL: Custom POSIX-Style Interactive Shell | *C++, Linux System Calls, Signals, termios, PTY Testing*

- Built a shell directly on Linux system calls (`fork/exec`, `pipe/dup2`, `sigaction`, `poll`, `waitpid`) with N-stage pipelines, ordered descriptor redirection and full job control (process groups, terminal handoff, Ctrl+Z, `fg/bg`), verified by a **138-check** suite including real-PTY keystroke tests.
- Made signal handling async-signal-safe: handlers only write one byte to a non-blocking self-pipe (preserving `errno`), and the main loop `poll`s it alongside the keyboard and reaps every child, eliminating race conditions and lost exit statuses.
- Cut per-keystroke cost from **∼25 syscalls to ∼3** (measured with `strace -c`) in a zero-dependency `termios` raw-mode line editor, and startup **52x** (5.2s to 0.1s) by reading file types from directory entries instead of a `stat()` per file.

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling worker threads and a fix for a Nagle/delayed-ACK stall in length-prefixed TCP framing cut 1GB single-peer downloads from **193s to under 13s**.
- Removed a data race and a file-wide lock by confining shared state to one mutex each and writing out-of-order chunks with lock-free positional `pwrite`, one writer per piece, after a SHA-1 check of every 512KB piece.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

RookDB: Vectorized Relational Storage Engine | *Rust, Buffer Management, B+Tree, Vectorized Execution*

- Co-architected a 4-crate SQL-99 RDBMS in Rust verified by **1,000+ tests**, building 8 KiB slotted pages, an on-disk B+Tree and one process-wide CLOCK buffer pool with pin counts and dirty write-back.
- Moved query operators from tuple-at-a-time virtual dispatch to 1,024-row batches with buffer reuse and made B+Tree lookups allocation-free by reading node pages in place; scans run at **20M rows/s** on 10M rows.
- Built a Grace hash join that spills to disk and recursively repartitions oversized partitions, joining **10M x 20M rows in 43 s** within a fixed **64 MiB** memory budget.

Warehouse Traffic Control: Fault-Tolerant Robot Coordination | *JavaScript, Web Workers*

- Designed a lease-and-fencing protocol coordinating 200 robots across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Implemented Chandy–Misra–Haas deadlock detection and crash recovery that rebuilds a manager's state in ∼4.3 s.
- Versus one central coordinator: **7.6x lower peak per-node load** (125 vs 950 msgs/s), and a region-manager crash cuts off **only 11 robots instead of all 120**.

Technical Skills

Languages: C/C++, Rust, Python, Bash, SQL, JavaScript.

Operating Systems & Linux: Processes & Job Control, Signals, Pipes & File Descriptors, Multithreading & Synchronization, termios, `/proc`, POSIX Sockets.

Memory & Performance: Buffer Pools & CLOCK Page Replacement, Page Layouts, Batched Execution, Syscall Profiling.

Tools: Make, Git, strace, ASan/UBSan, OpenSSL, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.